

Advanced (Habanero)

Q1. Solve the inequality $|x| > 3$.

- (A) $(-\infty, -3) \cup (3, \infty)$
 (B) $(-\infty, -3] \cup [3, \infty)$
 (C) $(-3, 3)$
 (D) $[-3, 3]$
 (E) $(-3, 3)$

Q2. Graph the solution set of $|x - 2| \leq 1$ on a number line.

- (A) $[1, 3]$
 (B) $(1, 3)$
 (C) $[2, 4]$
 (D) $(-1, 1)$
 (E) $(-2, 2)$

Q3. Solve $|x + 4| < 2$.

- (A) $(-6, -2)$
 (B) $(-2, 2)$
 (C) $[-6, -2]$
 (D) $(-2, 2)$
 (E) $[2, 6]$

Q4. The temperature in a town is never more than 25°C above or below 0°C . Express this as an inequality.

- (A) $|T| \leq 25$
 (B) $|T| \geq 25$
 (C) $T \leq 25$
 (D) $T \geq 25$
 (E) $T = 25$

Q5. Solve $|3x| > 12$.

- (A) $(-\infty, -4) \cup (4, \infty)$
 (B) $(-4, 4)$
 (C) $[-4, 4]$
 (D) $(-\infty, -4] \cup [4, \infty)$
 (E) $(-4, 4)$

Q6. A company packages coffee in bags that weigh between 500g and 600g. Express this as an inequality.

- (A) $|x - 550| \leq 50$
 (B) $|x - 500| \geq 100$
 (C) $|x - 600| \leq 50$
 (D) $x \geq 500$
 (E) $x \leq 600$

Q7. Solve $|x - 3| \geq 2$.

- (A) $(-\infty, 1] \cup [5, \infty)$
 (B) $(-\infty, 1) \cup (5, \infty)$
 (C) $[1, 5]$
 (D) $(-\infty, 1) \cup (5, \infty)$
 (E) $[1, 5]$

Q8. Graph the solution set of $|2x| < 6$.

- (A) $(-3, 3)$
 (B) $(-2, 2)$
 (C) $(-1, 1)$
 (D) $(-6, 6)$
 (E) $[-3, 3]$

Open Ended Questions (FRQ)

Q9. A bakery sells a total of 250 loaves of bread per day. They sell a combination of whole wheat and white bread. The number of whole wheat loaves sold is never more than 50 above or below the number of white bread loaves sold. Express this as an inequality and graph the solution set on a coordinate plane. Let x be the number of whole wheat loaves and y be the number of white bread loaves.**Q10.** The length of a rectangle is 5cm more than its width. The perimeter of the rectangle is never more than 50cm above or below 100cm. Express this as an inequality and find the possible values of the length and width of the rectangle. Let x be the width of the rectangle.

Chef's Tips

- Remind students to check their work carefully.
- Emphasize the importance of graphing the solution sets.
- Use real-world examples to illustrate the applications of absolute value inequalities.